

DAY-3 PANDAS

```
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Series
[2]: #1d array in Series
import pandas as pd
aspd.Series([1,2,3,4])
print(a)
# Series is a constructor
0    1
1    2
2    3
3    4
dtype: int64

[3]: import pandas as pd
aspd.Series([1,2,3,4])
print(a.values)
[1 2 3 4]

[4]: import pandas as pd
aspd.Series([1,2,3,4])
print(a.index)
RangeIndex(start=0, stop=4, step=1)

[5]: import pandas as pd
aspd.Series([1,2,3,4])
print(a.dtype)
int64

[6]: import pandas as pd
```

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[6]: import pandas as pd
aspd.Series([True,False,False,True])
print(a.dtype)
bool

[7]: import pandas as pd
aspd.Series([1,False,3,True])
print(a.dtype)
object

[8]: import pandas as pd
aspd.Series(['pavi','mugi','tm','pavin'])
print(a.dtype)
# since panda is a older version it is showing like object as output but the new version gives output as str only so dont get confuse....
object

[9]: #accessing elements
import pandas as pd
aspd.Series(['pavi','mugi','tm','pavin'])
print(a[1])
mugi

[12]: #accessing elements using loc
#diff btw index and loc [ when doing a slicing stop values is also included]
import pandas as pd
aspd.Series(['pavi','mugi','tm','pavin'])
print(a.loc[1])
mugi
```

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[13]: #diff btw index,loc and iloc [ when doing a slicing stop values is not included]
import pandas as pd
aspd.Series(['pavi','mugi','tm','pavin'])
print(a.iloc[1])
mugi

[14]: #slicing using index
import pandas as pd
aspd.Series(['pavi','mugi','tm','pavin'])
print(a[1:3])
1    mugi
2     tm
dtype: object

[15]: #slicing using loc
import pandas as pd
aspd.Series(['pavi','mugi','tm','pavin'])
print(a.loc[1:3])
### it prints the stop value also that is the index 3 here it print index 3 also here
1    mugi
2     tm
3    pavin
dtype: object

[17]: #slicing using loc
import pandas as pd
aspd.Series(['pavi','mugi','tm','pavin','gayuma','george'])
print(a.iloc[2:5])
2     tm
3    pavin
4    gayuma
5    george
```

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```
[18]: import pandas as pd
arrsp.Series([1,2,3,4,5],index=['A','B','C','D','E'])
print(arr)
# since it is a labelled array i can label my index as i wish...
A    1
B    2
C    3
D    4
E    5
dtype: int64

[20]: import pandas as pd
arrsp.Series([1,2,3,4,5],index=['A','B','C','D','E'])
print(arr['E'])
5

[21]: import pandas as pd
arrsp.Series([1,2,3,4,5],index=['A','B','C','D','E'])
print(arr*5)
#Vectorized operation does not need any Looping..
A    6
B    7
C    8
D    9
E   10
dtype: int64

[23]: import pandas as pd
arrsp.Series([1,2,3,4,5],index=['A','B','C','D','E'])
print(arr.iloc[1])
# can be accessed only with position indexing Labelled indexing doesnt work here
```

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```
2

[25]: import pandas as pd
arrsp.Series([1,2,3,4,5],index=['A','B','C','D','E'])
print(arr.loc['A'])
# can be accessed using labelled values also
1

[26]: #statistical function
import pandas as pd
arrsp.Series([1,2,3,4,5],index=['A','B','C','D','E'])
print(arr.sum())
15

[27]: import pandas as pd
arrsp.Series([1,2,3,4,5],index=['A','B','C','D','E'])
print(arr.std())
1.5811388300841898

[28]: import pandas as pd
arrsp.Series([1,2,3,4,5],index=['A','B','C','D','E'])
print(arr.max())
5

[29]: import pandas as pd
arrsp.Series([1,2,3,4,5],index=['A','B','C','D','E'])
print(arr.min())
1

[30]: import pandas as pd
#arrsp.Series([1,2,3,4,5],index=['A','B','C','D','E'])
```

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```
[30]: import pandas as pd
arrsp.Series([1,2,3,4,5],index=['A','B','C','D','E'])
print(arr.median())
3.0

[31]: import pandas as pd
arrsp.Series([1,2,3,4,5],index=['A','B','C','D','E'])
print(arr.mean())
3.0

[32]: #boolean indexing
import pandas as pd
arrsp.Series([13,42,32,54,52],index=['A','B','C','D','E'])
print(arr[arr>30])
B    42
C    32
D    54
E    52
dtype: int64

[34]: #count values
import pandas as pd
arrsp.Series([13,42,32,54,52],index=['A','B','C','D','E'])
print(len(arr))
5

[35]: #it return individual values count
import pandas as pd
arrsp.Series([13,42,32,54,52],index=['A','B','C','D','E'])
print(arr.value_counts())
```

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13 1
42 1
32 1
54 1
52 1
Name: count, dtype: int64

[37]: #it return individual values count
import pandas as pd
arr=arrpd.Series([13,42,32,42,54,52],index=['A','B','C','D','E','F'])
print(arr.value_counts())

42 2
13 1
32 1
54 1
52 1
Name: count, dtype: int64

[38]: #string operation
name=arrpd.Series(['jack','nisha','vishnu','rithesh','sree'])
print(name.str.upper())

0 JACK
1 NISHA
2 VISHNU
3 RITESH
4 SREE
dtype: object

[39]: #string operation
name=arrpd.Series(['jack','nisha','vishnu','rithesh','sree'])
print(name.str.len())

0 4
```

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0 4
1 5
2 6
3 7
4 4
dtype: int64

[40]: #Attributes
import pandas as pd
s = pd.Series([10, 20, 30])
print(s.size)
print(s.shape)
print(s.dtype)

3
(3,)
int64

[43]: import pandas as pd
d = pd.Series([
    "Math": 90,
    "Science": 85,
    "English": 95])
print(d)
#values 0 la start agura mari kudutha it does take it bcz it considers it as hexadecimal values

Math 90
Science 85
English 95
dtype: int64

[45]: s1=arrpd.Series([10,20,30],index=['A','B','C'])
s2=arrpd.Series([1,2,3],index=['B','C','D'])
print(s1+s2)
# since A and D index is not there in both of the list so it return null value
```

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# since A and D index is not there in both of the list so it return null value
# why the dtype is float here....IT CONSIDERS NULL AS FLOAT VALUE so every thing is changed to float values

A NaN
B 21.0
C 32.0
D NaN
dtype: float64

[46]: s=arrpd.Series([1,2,3,4,5])
print(s.count())

5

[48]: s=arrpd.Series([1,2,2,4,5,3,4,5])
print(s.unique())

[1 2 4 5 3]

[50]: s=arrpd.Series([1,2,2,4,5,3,4,5])
print(s.nunique())
#represents the count of unique values

5

[51]: #string operation
name=arrpd.Series(['jack','nisha','vishnu','rithesh','sree'])
print(name.str.capitalize())

0 Jack
1 Nisha
2 Vishnu
3 Rithesh
4 Sree
dtype: object
```

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```
[54]: import pandas as pd
arr=pd.Series([13,42,32,42,54,52],index=['A','B','C','D','E','F'])
print(arr[['A','D','E','C']])
# can access both using labelled and index value
```

A 13
D 42
E 54
C 32
dtype: int64

Data frame

data frame can be created in 3 ways

1.using dictionary

```
[62]: import pandas as pd
data={
    "name":["monika","nagul","prem"],
    "rollno":[43,56,34],
    "age":[12,14,25]}
df=pd.DataFrame(data)
print(df)
```

	name	rollno	age
0	monika	43	12
1	nagul	56	14
2	prem	34	25

2.using series

```
[66]: s1=pd.Series([1,2,3,4,5])
```

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3.using series

```
[65]: s1=pd.Series([1,2,3,4,5])
s2=pd.Series([2.5,5.0,1.5,6.7,8.9])
s3=pd.Series([True,False,False,True])
df=pd.DataFrame({'sno':s1,
                 'cgpa':s2,
                 'result':s3})
print(df)
```

	sno	cgpa	result
0	1	2.5	True
1	2	5.0	False
2	3	1.5	False
3	4	6.7	True
4	5	8.9	NaN

3.list of list

```
[67]: data=[['athish',22,98],['rithika',20,89],['pratheksha',21,87]]
df=pd.DataFrame(data,columns=['name','age','marks'])
print(df)
```

	name	age	marks
0	athish	22	98
1	rithika	20	89
2	pratheksha	21	87

4.list of dictionaries

```
[71]: data=[
    {'name':"mugl","course":"data science","grade":"A"},
    {'name':"thiru","course":"cs","grade":"A+"},
    {'name':"nivas","course":"cs","grade":"B"}]
df=pd.DataFrame(data,index=[0,1,2],columns=["name","age","marks"])
```

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```
df=pd.DataFrame(data,index=["std1","std2","std3"])
print(df)
```

	name	course	grade
std1	mugl	data science	A
std2	thiru	cs	A+
std3	nivas	cs	B

[73]: # creating 1 extra column to it

```
data=[
    {'name':"mugl","course":"data science","grade":"A"},
    {'name':"thiru","course":"cs","grade":"A+"},
    {'name':"nivas","course":"cs","grade":"B"}]
df=pd.DataFrame(data,index=["std1","std2","std3"])
df['marks']= [98,87,54]
print(df)
```

	name	course	grade	marks
std1	mugl	data science	A	98
std2	thiru	cs	A+	87
std3	nivas	cs	B	54

[96]: # creating 1 extra row to it

```
data=[
    {'name':"mugl","course":"data science","grade":"A"},
    {'name':"thiru","course":"cs","grade":"A+"},
    {'name':"nivas","course":"cs","grade":"B"}]
df=pd.DataFrame(data,index=["std1","std2","std3"])
df.loc["std4"]=["keerthu","csa","B"]
print(df)
```

#another way using concatenation method

```
data=[
    {'name':"mugl","course":"data science","grade":"A"},
```

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```

#another way using concatenation method
data=[
    {"name":"mugi","course":"data science","grade":"A"},
    {"name":"thiru","course":"cs","grade":"A+"},
    {"name":"nivas","course":"cs","grade":"B"}]
dfspd.DataFrame(data,index=["std1","std2","std3"])
std4spd.DataFrame([{"name":"mahi","course":"ds","grade":"o"}],index=["std4"])
dfspd.concat([df,std4])
print(df)

# to create multiple rows
studentspd.DataFrame([{"name":"sanjai","course":"csda","grade":"O+","marks":98},
    {"name":"meena","course":"cs","grade":"A","marks":54}],index=["std5","std6"])
dfspd.concat([df,students])
print(df)

```

	name	course	grade
std1	mugi	data science	A
std2	thiru	cs	A+
std3	nivas	cs	B
std4	keerthu	csa	B

	name	course	grade
std1	mugi	data science	A
std2	thiru	cs	A+
std3	nivas	cs	B
std4	mahi	ds	o

	name	course	grade	marks
std1	mugi	data science	A	NaN
std2	thiru	cs	A+	NaN
std3	nivas	cs	B	NaN
std4	mahi	ds	o	NaN

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```

std1    name    course grade
std2    thiru   cs      A+
std3    nivas   cs      B
std4    keerthu csa     B
std1    name    course grade
std2    thiru   cs      A+
std3    nivas   cs      B
std4    mahi    ds      o
std1    name    course grade marks
std2    thiru   cs      A+   NaN
std3    nivas   cs      B    NaN
std4    mahi    ds      o    NaN
std5    sanjai  csda     O+   98.0
std6    meena   cs      A    54.0
std1    mugi
std2    thiru
std3    nivas
std4    mahi
std5    sanjai
std6    meena
Name: name, dtype: object

```

[98]: #column Access by bracket notation and dot notation
dot notation

```

data=[
    {"name":"mugi","course":"data science","grade":"A"},
    {"name":"thiru","course":"cs","grade":"A+"},
    {"name":"nivas","course":"cs","grade":"B"}]
dfspd.DataFrame(data,index=["std1","std2","std3"])
df["marks"]= [98,87,54]
print(df)
print(df.course)

```

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```

##bracket notation
data=[
    {"name":"mugi","course":"data science","grade":"A"},
    {"name":"thiru","course":"cs","grade":"A+"},
    {"name":"nivas","course":"cs","grade":"B"}]
dfspd.DataFrame(data,index=["std1","std2","std3"])
df["marks"]= [98,87,54]
print(df["name"])
print(df[["grade","marks"]])

std1    data science
std2    cs
std3    cs
Name: course, dtype: object
std1    mugi
std2    thiru
std3    nivas
Name: name, dtype: object
std1    grade marks
std1    A      98
std2    A+    87
std3    B      54

```

[8]: #shape attribute

```

import pandas as pd
data=[
    {"name":"mugi","course":"data science","grade":"A"},
    {"name":"thiru","course":"cs","grade":"A+"},
    {"name":"nivas","course":"cs","grade":"B"}]
dfspd.DataFrame(data,index=["std1","std2","std3"])
df["marks"]= [98,87,54]
std4spd.DataFrame([{"name":"mahi","course":"ds","grade":"o"}],index=["std4"])
dfspd.concat([df,std4])

```

```
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dfspd.concat((df,std4))

studentspd.DataFrame([{"name":"sanjai","course":"csda","grade":"O+","marks":98},
                      {"name":"meena","course":"cs","grade":"A","marks":54}],index=["std5","std6"])
dfspd.concat((df,students))
print(df.shape)

(6, 4)

[9]: #find no of row
import pandas as pd
data=[
    {"name":"mugi","course":"data science","grade":"A"},
    {"name":"thiru","course":"cs","grade":"A+"},
    {"name":"nivas","course":"cs","grade":"B"}]
dfspd.DataFrame(data,index=["std1","std2","std3"])
df["marks"]=[98,87,54]
std4spd.DataFrame([{"name":"mahi","course":"ds","grade":"o"}],index=["std4"])
dfspd.concat((df,std4))

studentspd.DataFrame([{"name":"sanjai","course":"csda","grade":"O+","marks":98},
                      {"name":"meena","course":"cs","grade":"A","marks":54}],index=["std5","std6"])
dfspd.concat((df,students))
print(df.shape)
print(df.shape[0])

(6, 4)
6

[10]: #find no of columns
import pandas as pd
data=[
    {"name":"mugi","course":"data science","grade":"A"},
    {"name":"thiru","course":"cs","grade":"A+"},
```

```
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[11]: #find no of columns
import pandas as pd
data=[
    {"name":"mugi","course":"data science","grade":"A"},
    {"name":"thiru","course":"cs","grade":"A+"},
    {"name":"nivas","course":"cs","grade":"B"}]
dfspd.DataFrame(data,index=["std1","std2","std3"])
df["marks"]=[98,87,54]
std4spd.DataFrame([{"name":"mahi","course":"ds","grade":"o"}],index=["std4"])
dfspd.concat((df,std4))

studentspd.DataFrame([{"name":"sanjai","course":"csda","grade":"O+","marks":98},
                      {"name":"meena","course":"cs","grade":"A","marks":54}],index=["std5","std6"])
dfspd.concat((df,students))
print(df.shape)
print(df.shape[1])

(6, 4)
4

[17]: #dtype attribute
#used to check datatype of all columns
import pandas as pd
data=[
    {"name":"mugi","course":"data science","grade":"A"},
    {"name":"thiru","course":"cs","grade":"A+"},
    {"name":"nivas","course":"cs","grade":"B"}]
dfspd.DataFrame(data,index=["std1","std2","std3"])
df["marks"]=[98,87,54]
std4spd.DataFrame([{"name":"mahi","course":"ds","grade":"o"}],index=["std4"])
dfspd.concat((df,std4))

studentspd.DataFrame([{"name":"sanjai","course":"csda","grade":"O+","marks":98},
```

```
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{"name":"meena","course":"cs","grade":"A","marks":54}],index=["std5","std6"])
dfspd.concat((df,students))

print(df.dtypes)

name      object
course    object
grade     object
marks    float64
dtype: object

[16]: #info
import pandas as pd
data=[
    {"name":"mugi","course":"data science","grade":"A"},
    {"name":"thiru","course":"cs","grade":"A+"},
    {"name":"nivas","course":"cs","grade":"B"}]
dfspd.DataFrame(data,index=["std1","std2","std3"])
df["marks"]=[98,87,54]
std4spd.DataFrame([{"name":"mahi","course":"ds","grade":"o"}],index=["std4"])
dfspd.concat((df,std4))

studentspd.DataFrame([{"name":"sanjai","course":"csda","grade":"O+","marks":98},
                      {"name":"meena","course":"cs","grade":"A","marks":54}],index=["std5","std6"])
dfspd.concat((df,students))

print(df.info())

<class 'pandas.core.frame.DataFrame'>
Int64Index: 6 entries, std1 to std6
Data columns (total 4 columns):
 #   Column  Non-Null Count  Dtype
---  -
 0   name    6 non-null        object
```

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```
name      non-null      object
1  course  6 non-null  object
2  grade   6 non-null  object
3  marks   5 non-null  float64
dtypes: float64(1), object(3)
memory usage: 240.0+ bytes
None
```

```
[15]: import pandas as pd
data=[
    {"name":"mugi","course":"data science","grade":"A"},
    {"name":"thiru","course":"cs","grade":"A+"},
    {"name":"nivas","course":"cs","grade":"B"}]
dfpd.DataFrame(data,index=["std1","std2","std3"])
df['marks']=98,87,54
std4spd.DataFrame([{"name":"mahi","course":"ds","grade":"o"}],index=["std4"])
dfpd.concat((df,std4))

studentsspd.DataFrame([{"name":"sanjai","course":"csda","grade":"O+","marks":98},
                        {"name":"meena","course":"cs","grade":"A","marks":54}],index=["std5","std6"])
dfpd.concat((df,students))

print(df.describe())
```

	marks
count	5.000000
mean	78.200000
std	22.543252
min	54.000000
25%	54.000000
50%	87.000000
75%	98.000000
max	98.000000

```
[20]: #select specific row
```

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```
[20]: #select specific row
import pandas as pd
data=[
    {"name":"mugi","course":"data science","grade":"A"},
    {"name":"thiru","course":"cs","grade":"A+"},
    {"name":"nivas","course":"cs","grade":"B"}]
dfpd.DataFrame(data,index=["std1","std2","std3"])
df['marks']=98,87,54
std4spd.DataFrame([{"name":"mahi","course":"ds","grade":"o"}],index=["std4"])
dfpd.concat((df,std4))

studentsspd.DataFrame([{"name":"sanjai","course":"csda","grade":"O+","marks":98},
                        {"name":"meena","course":"cs","grade":"A","marks":54}],index=["std5","std6"])
dfpd.concat((df,students))

print(df.loc['std4'])
```

	name	course	grade	marks
	mahi	ds	o	NaN

Name: std4, dtype: object

```
[21]: import pandas as pd
data=[
    {"name":"mugi","course":"data science","grade":"A"},
    {"name":"thiru","course":"cs","grade":"A+"},
    {"name":"nivas","course":"cs","grade":"B"}]
dfpd.DataFrame(data,index=["std1","std2","std3"])
df['marks']=98,87,54
std4spd.DataFrame([{"name":"mahi","course":"ds","grade":"o"}],index=["std4"])
dfpd.concat((df,std4))

studentsspd.DataFrame([{"name":"sanjai","course":"csda","grade":"O+","marks":98},
                        {"name":"meena","course":"cs","grade":"A","marks":54}],index=["std5","std6"])
dfpd.concat((df,students))

print(df.iloc[3])
```

	name	course	grade	marks
	mahi	ds	o	NaN

Name: std4, dtype: object

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```
studentsspd.DataFrame([{"name":"sanjai","course":"csda","grade":"O+","marks":98},
                        {"name":"meena","course":"cs","grade":"A","marks":54}],index=["std5","std6"])
dfpd.concat((df,students))

print(df.iloc[3])
```

	name	course	grade	marks
	mahi	ds	o	NaN

Name: std4, dtype: object

```
[23]: #updatevalue
import pandas as pd
data=[
    {"name":"mugi","course":"data science","grade":"A"},
    {"name":"thiru","course":"cs","grade":"A+"},
    {"name":"nivas","course":"cs","grade":"B"}]
dfpd.DataFrame(data,index=["std1","std2","std3"])
df['marks']=98,87,54
std4spd.DataFrame([{"name":"mahi","course":"ds","grade":"o"}],index=["std4"])
dfpd.concat((df,std4))

studentsspd.DataFrame([{"name":"sanjai","course":"csda","grade":"O+","marks":98},
                        {"name":"meena","course":"cs","grade":"A","marks":54}],index=["std5","std6"])
dfpd.concat((df,students))
df.loc['std3','grade']='B++'
print(df)
```

	name	course	grade	marks
std1	mugi	data science	A	98.0
std2	thiru	cs	A+	87.0
std3	nivas	cs	B++	54.0
std4	mahi	ds	o	NaN

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JupyterLab Python [conda env:base] *  

```
[34]: # to delete a column
import pandas as pd
data=[
    {"name":"mugli","course":"data science","grade":"A"},
    {"name":"thiru","course":"cs","grade":"A+"},
    {"name":"nivas","course":"cs","grade":"B"}]
dfapd.DataFrame(data,index=["std1","std2","std3"])
df['marks']=98,87,54
std4spd.DataFrame([{"name":"mahi","course":"ds","grade":"o"}],index=["std4"])
dfapd.concat((df,std4))

studentspd.DataFrame([{"name":"sanjai","course":"csda","grade":"O+","marks":98},
    {"name":"meena","course":"cs","grade":"A","marks":54}],index=["std5","std6"])
dfapd.concat((df,students))

df.drop("grade",axis=1,inplace=True)
print(df)
```

	name	course	marks
std1	mugli	data science	98.0
std2	thiru	cs	87.0
std3	nivas	cs	54.0
std4	mahi	ds	NaN
std5	sanjai	csda	98.0
std6	meena	cs	54.0

```
[33]: #to del row
import pandas as pd
data=[
    {"name":"mugli","course":"data science","grade":"A"},
    {"name":"thiru","course":"cs","grade":"A+"},
```

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JupyterLab Python [conda env:base] *  

```
[35]: #sorting
import pandas as pd
data=[
    {"name":"mugli","course":"data science","grade":"A"},
    {"name":"thiru","course":"cs","grade":"A+"},
    {"name":"nivas","course":"cs","grade":"B"}]
dfapd.DataFrame(data,index=["std1","std2","std3"])
df['marks']=98,87,54
std4spd.DataFrame([{"name":"mahi","course":"ds","grade":"o"}],index=["std4"])
dfapd.concat((df,std4))

studentspd.DataFrame([{"name":"sanjai","course":"csda","grade":"O+","marks":98},
    {"name":"meena","course":"cs","grade":"A","marks":54}],index=["std5","std6"])
dfadf.concat((df,students))

dfadf.drop("std6",axis=0)
print(df)
```

	name	course	grade	marks
std1	mugli	data science	A	98.0
std2	thiru	cs	A+	87.0
std3	nivas	cs	B	54.0
std4	mahi	ds	o	NaN
std5	sanjai	csda	O+	98.0

```
[35]: #sorting
import pandas as pd
data=[
    {"name":"mugli","course":"data science","grade":"A"},
    {"name":"thiru","course":"cs","grade":"A+"},
    {"name":"nivas","course":"cs","grade":"B"}]
dfapd.DataFrame(data,index=["std1","std2","std3"])
df['marks']=98,87,54
std4spd.DataFrame([{"name":"mahi","course":"ds","grade":"o"}],index=["std4"])
dfapd.concat((df,std4))

studentspd.DataFrame([{"name":"sanjai","course":"csda","grade":"O+","marks":98},
    {"name":"meena","course":"cs","grade":"A","marks":54}],index=["std5","std6"])
```



```
dfapd.concat((df,students))
df.sort_values('marks',inplace=True)
print(df)
### marks ascending order
```

	name	course	grade	marks
std3	nivas	cs	B	54.0
std6	meena	cs	A	54.0
std2	thiru	cs	A+	87.0
std1	mugi	data science	A	98.0
std5	sanjai	csda	O+	98.0
std4	mahi	ds	o	NaN

```
[37]: import pandas as pd
data=[
{"name":"mugi","course":"data science","grade":"A"},
{"name":"thiru","course":"cs","grade":"A+"},
{"name":"nivas","course":"cs","grade":"B"}]
dfapd.DataFrame(data,index=["std1","std2","std3"])
df['marks']=98,87,54
std4spd.DataFrame([{"name":"mahi","course":"ds","grade":"o"}],index=["std4"])
dfapd.concat((df,std4))

studentspd.DataFrame([{"name":"sanjai","course":"csda","grade":"O+","marks":98},
{"name":"meena","course":"cs","grade":"A","marks":54}],index=["std5","std6"])
dfapd.concat((df,students))
df.sort_values('marks',ascending=False,inplace=True)
print(df)
### marks descending order
```

	name	course	grade	marks
std1	mugi	data science	A	98.0
std5	sanjai	csda	O+	98.0